

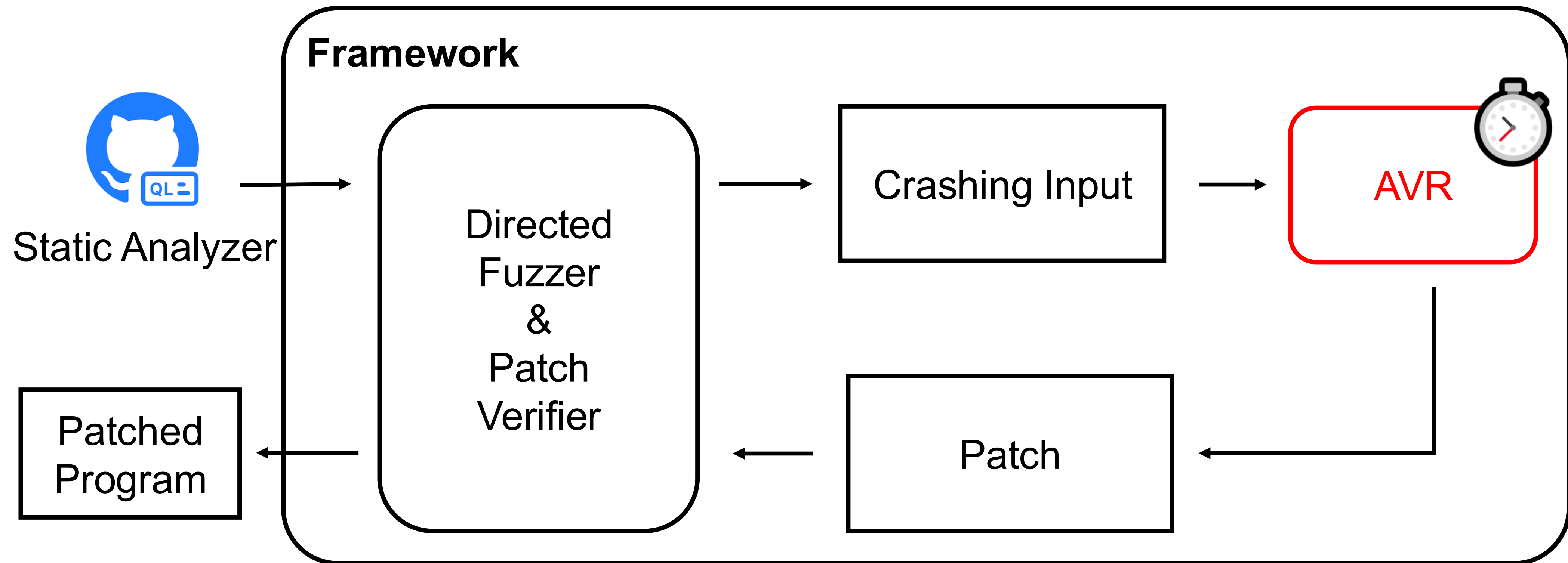
Accelerating Automated Vulnerability Repair with Meta Program

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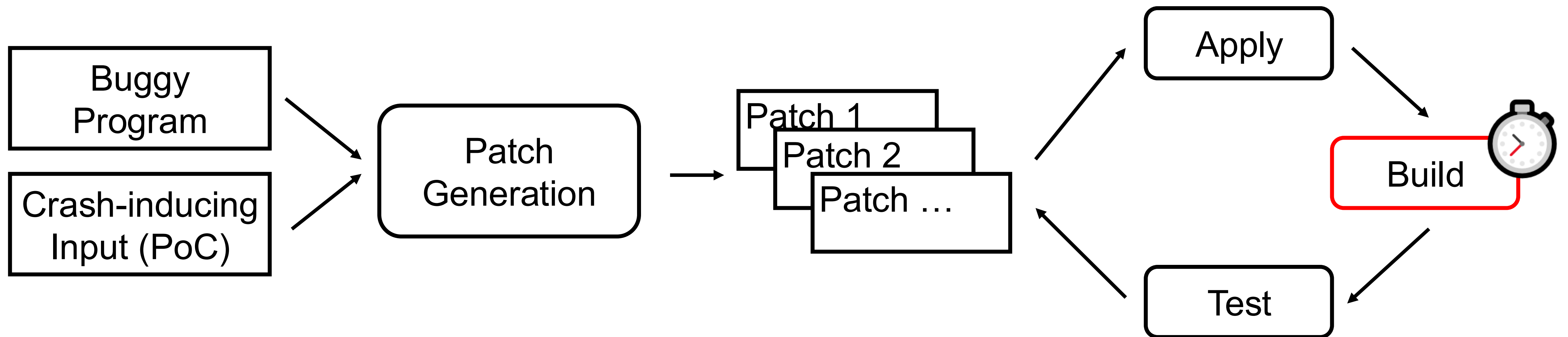
LOFT@UNIST

In previous workshop...

- Integrating Static analysis, Directed fuzzing, AVR and Patch verification

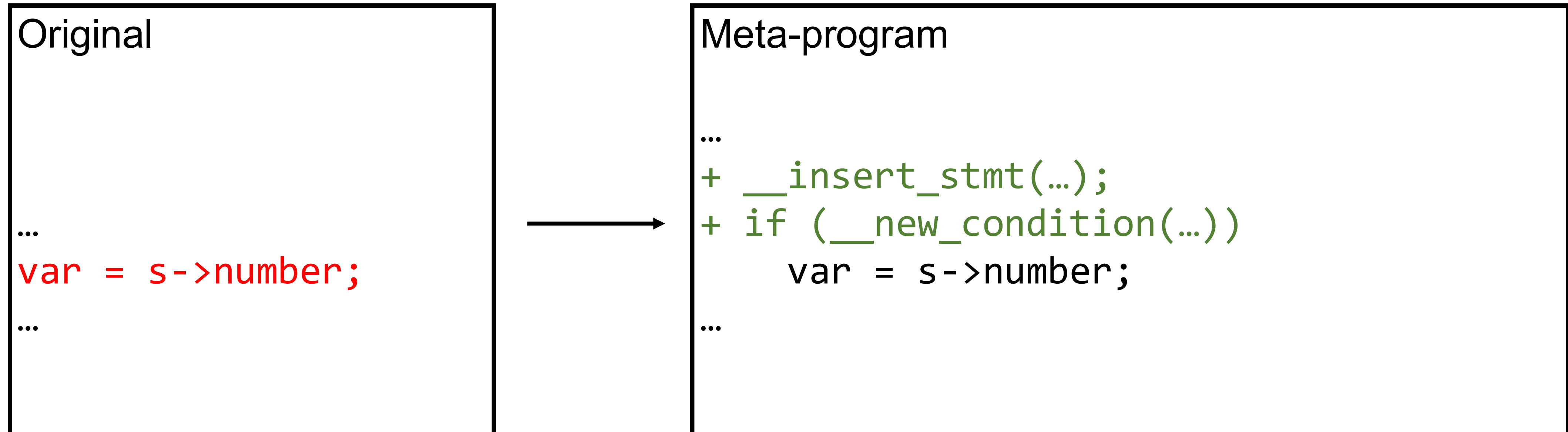


Automated Vulnerability Repair (AVR)



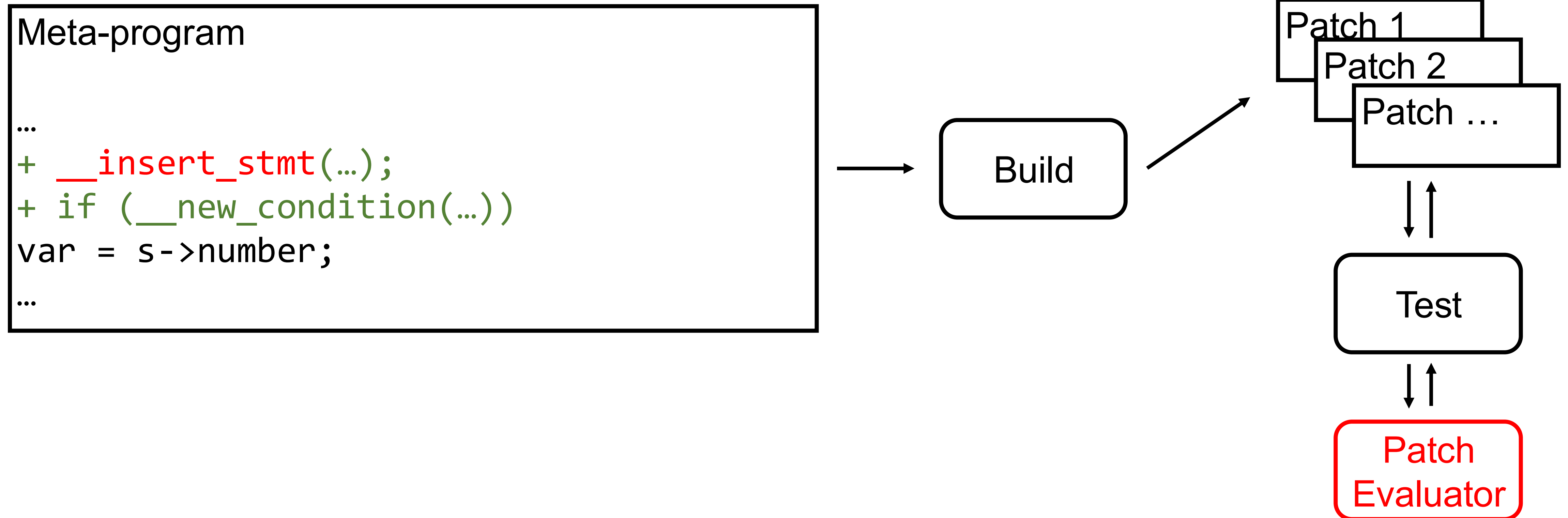
Meta Program

- Apply every patch templates to single source code



Meta Program

- Build program only once



Simple C Interpreter

- Give patch with environment variable

```
if (s == NULL) return;
```



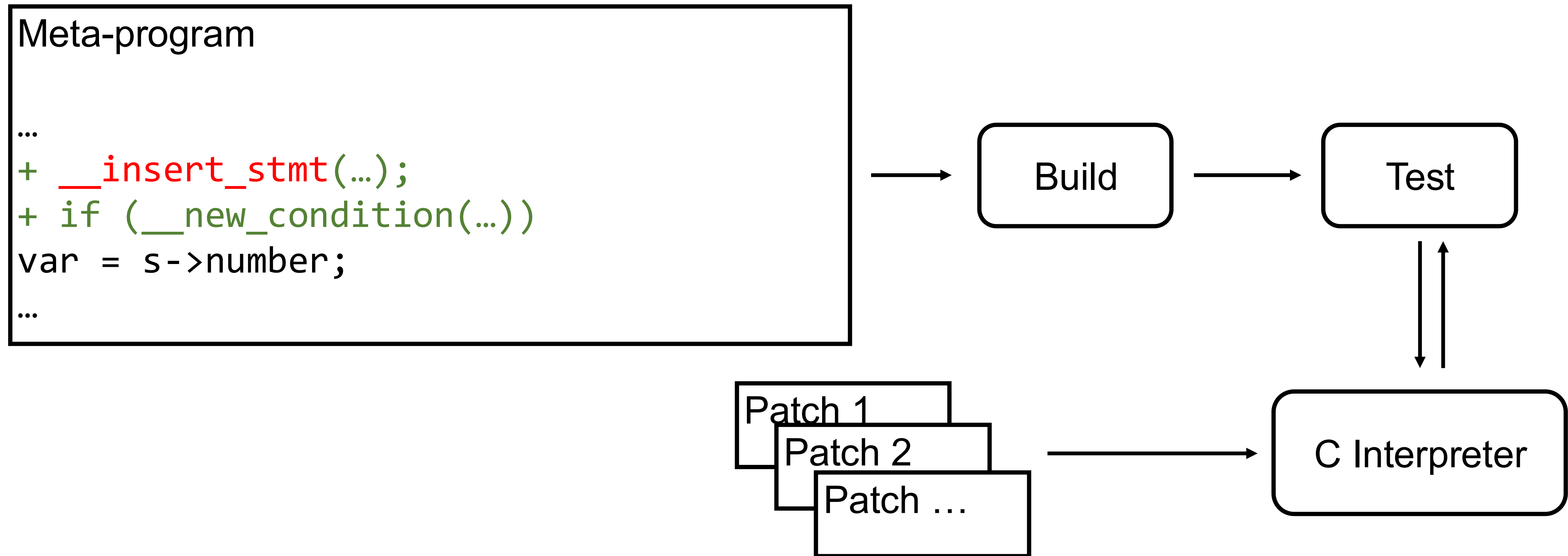
```
$ export INSERT_STMT='...'  
$ ./binary input
```



C Interpreter

```
patch = getenv("INSERT_STMT");  
evaluate(patch);
```

Summary



Evaluation

- Tiny benchmarks: 18 bugs
- w/o meta program: **6.52s** vs. w/ meta program: **0.87s** (86.63%)